STRATUM · CASE STUDY

# Procure-to-Pay Intelligence

What 1.6 million ERP events reveal about how a company really buys and pays.

Process mining · Compliance analytics · Causal inference · Machine learning

THE BRIEF

# A €997M order book and four hard questions

The process owner has compliance questions, and the finance team needs to plan. The answers sit in **1,595,923 ERP events** spread across 76,349 purchase orders from 1,975 vendors.



## What really happens?

The official process and the recorded one are rarely the same.



## Where do controls fail?

And how much money sits behind each failure?



## When does cash go out?

Payment dates and near-term outflow, with honest uncertainty.



## What actually causes delay?

Separating real causes from things that merely travel with them.

THE ANSWERS

# What the analysis found

**€200.2M**

CONTROLS

of order value carries at least one control exception (6.7% of lines).

**−0.1 days**

REWORK

is the adjusted effect of order changes on payment. The naive gap of 5.1 days disappears.

**3.6%**

CASH

average miss on 30-day cash-out in three backtests, against 5.9% for a naive repeat.

**83 days**

TIMING

median time to pay when open items count. Looking only at paid items gives 77.

KNOW YOUR DATA

# The field that would have broken everything

SUMMED NAIVELY

**€5.7B**

apparent goods and invoice value

TRUE ORDER BOOK

**€997M**

total order value

## A name that misleads

- “Cumulative net worth” repeats the line's running value on every event, so it isn't the amount of any one document.
- A first version summed it, and the totals didn't reconcile to the order book. The check caught it.
- Value-based matching isn't possible here, so controls test document counts and exposure uses order value.

ENGINEERING

# An ERP data model you can trust

## Bronze

Streamed, untouched

- 730 MB XES log parsed in about 30 seconds
- Checksum verified on every run
- External XML entities disabled

## Staging

dbt on DuckDB

- Typed events and order lines
- Implausible dates flagged, not dropped
- Every event numbered in process order

## Marts

Star schema

- Document, line and event grains
- Vendor, user, activity and date dimensions
- Compliance, variants and process graph

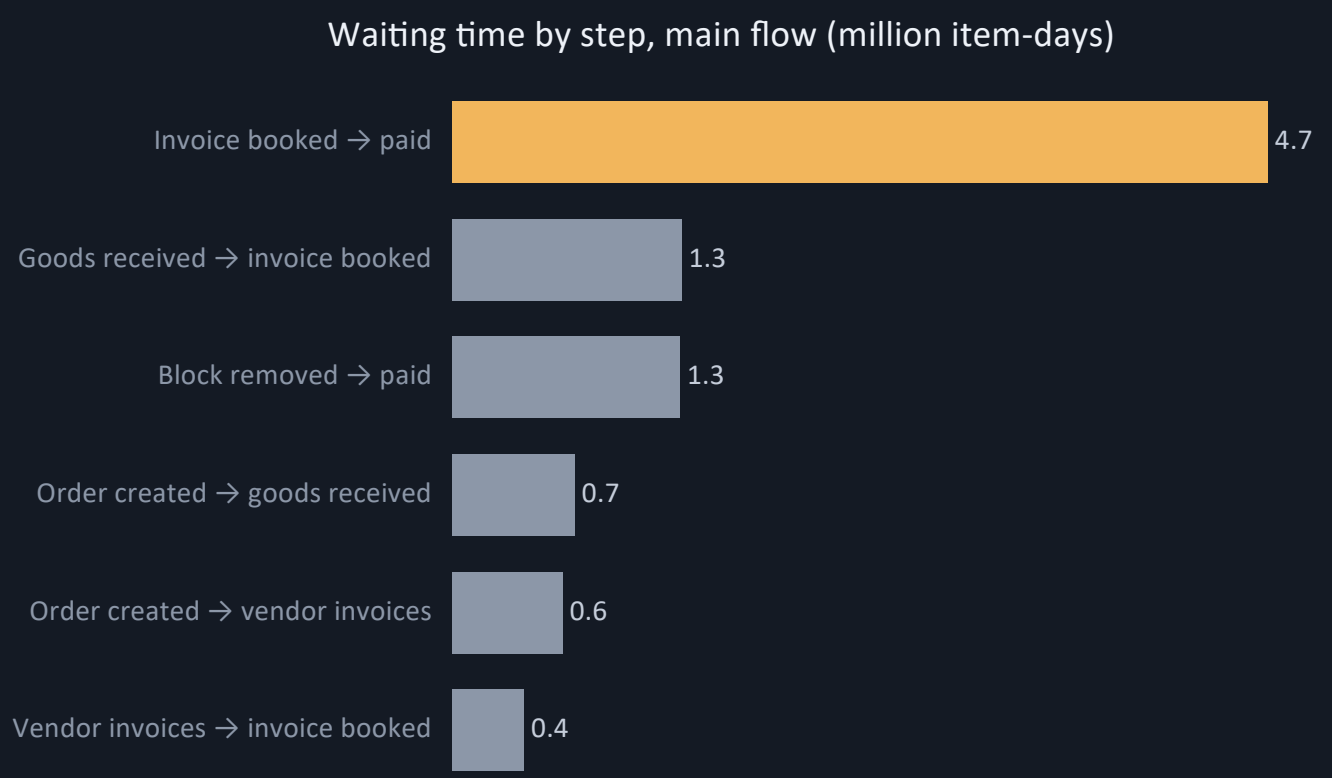
**16** dbt models

**31 / 31** dbt tests pass, including a rule unit test

**1.6M** events reconciled to source

PROCESS MINING

# Most waiting happens after the invoice is booked




**84%**

of paid lines in the main flow ran straight through

**0.96**

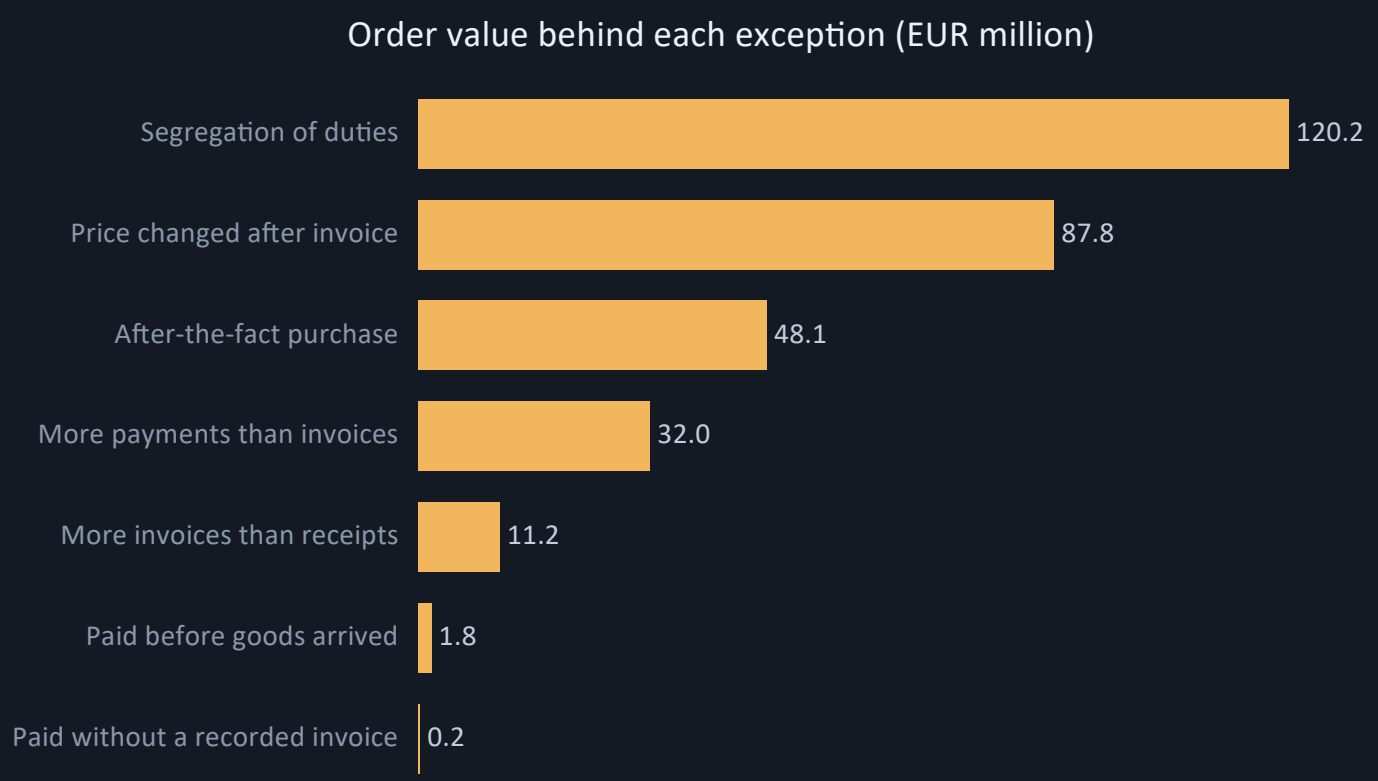
fitness of the discovered model on the main flow

**0.68**

on the GR-based flow: far less disciplined

COMPLIANCE

# €200.2M of orders carry a control exception




**7,551**

lines with one person on both sides of a control

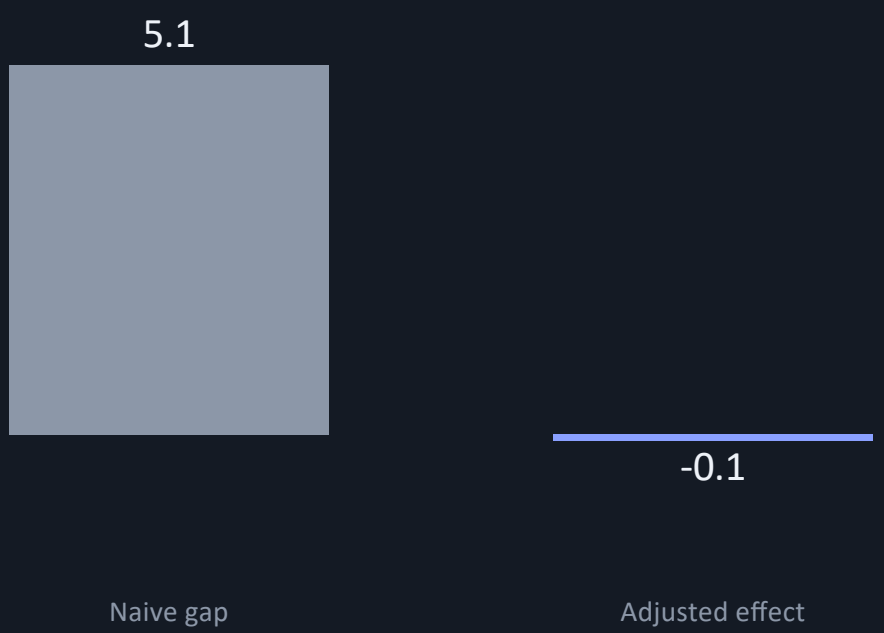
**3 of 10**

controls recorded no exceptions at all, and goods receivers never pay

*An exception is a prompt to review, not proof of wrongdoing. Exposure is the order value of the line.*

CAUSAL INFERENCE

# Reworked orders look slower. Rework isn't why




## Three traps, handled

**Confounding:** double machine learning adjusts for vendor, value and spend area.

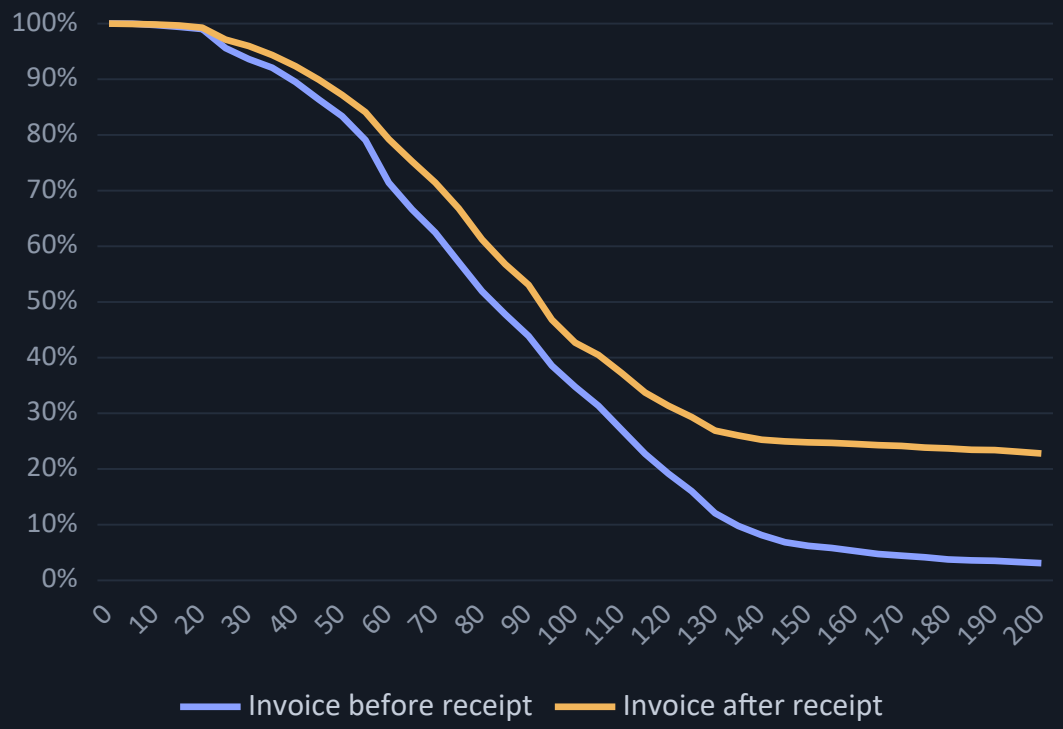
**Immortal time:** only orders still unpaid at day 14 count, with changes inside those 14 days.

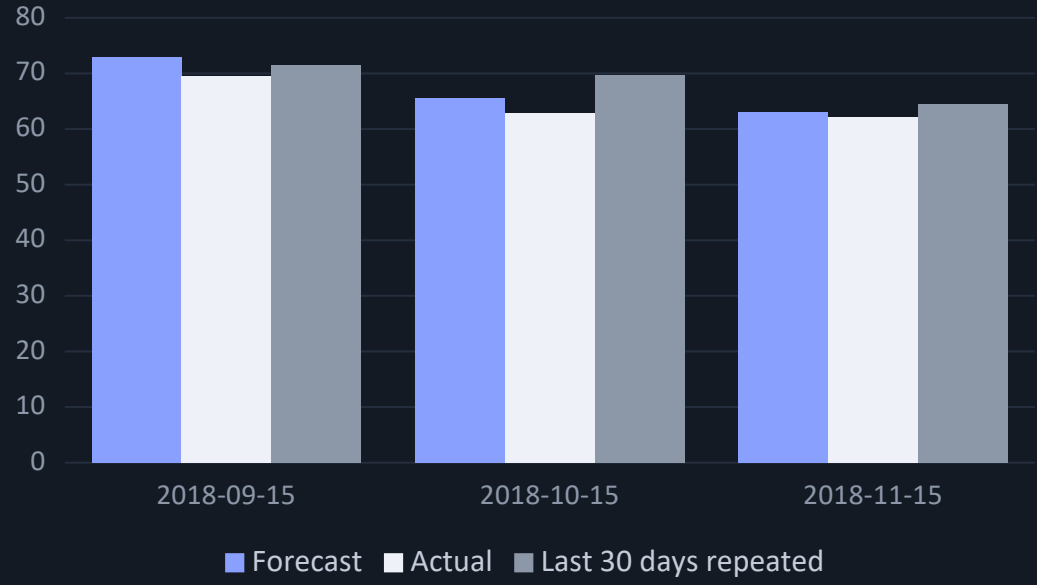
**Censoring:** the outcome is unpaid days in a fixed 180-day window.

**Result:** 95% CI -1.6 to +1.4 days on 108,383 lines. Placebo: +0.5.

SURVIVAL ANALYSIS

# Forecasting cash from the open book




Next 30 days: about €67.7M expected to go out. Average backtest miss 3.6% vs 5.9% naive.

HIERARCHICAL BAYES

# Finding vendors that are truly a problem

**1**

median lines per vendor in the raw top 20: league tables crown one-order flukes

**55**

median lines in the Bayesian top 20: real volume, worth a conversation

**1 of 20**

vendors appear on both lists, so the choice of method changes who you call

| Predicting each vendor's second half | Log score (higher is better) | Weighted error |
|---|---|---|
| **Hierarchical Bayes** | **-0.224** | **0.061** |
| Raw vendor rate | -0.294 | 0.064 |
| Overall rate | -0.240 | 0.077 |

UNCERTAINTY AND LIVE PREDICTION

# Predictions you can rely on

## Conformal prediction

Does a “90% range” really hold 90% of the time?

**79%**

coverage from the raw quantile model

**92%**

after conformal calibration

*Coverage on the smaller flow reaches 88%, and ranges are 105 days wide: honest rather than impressive.*

## Deep sequence model

A PyTorch GRU reads events as they happen.

**19.8 d**

error on time remaining, vs 24.2 days for the baseline

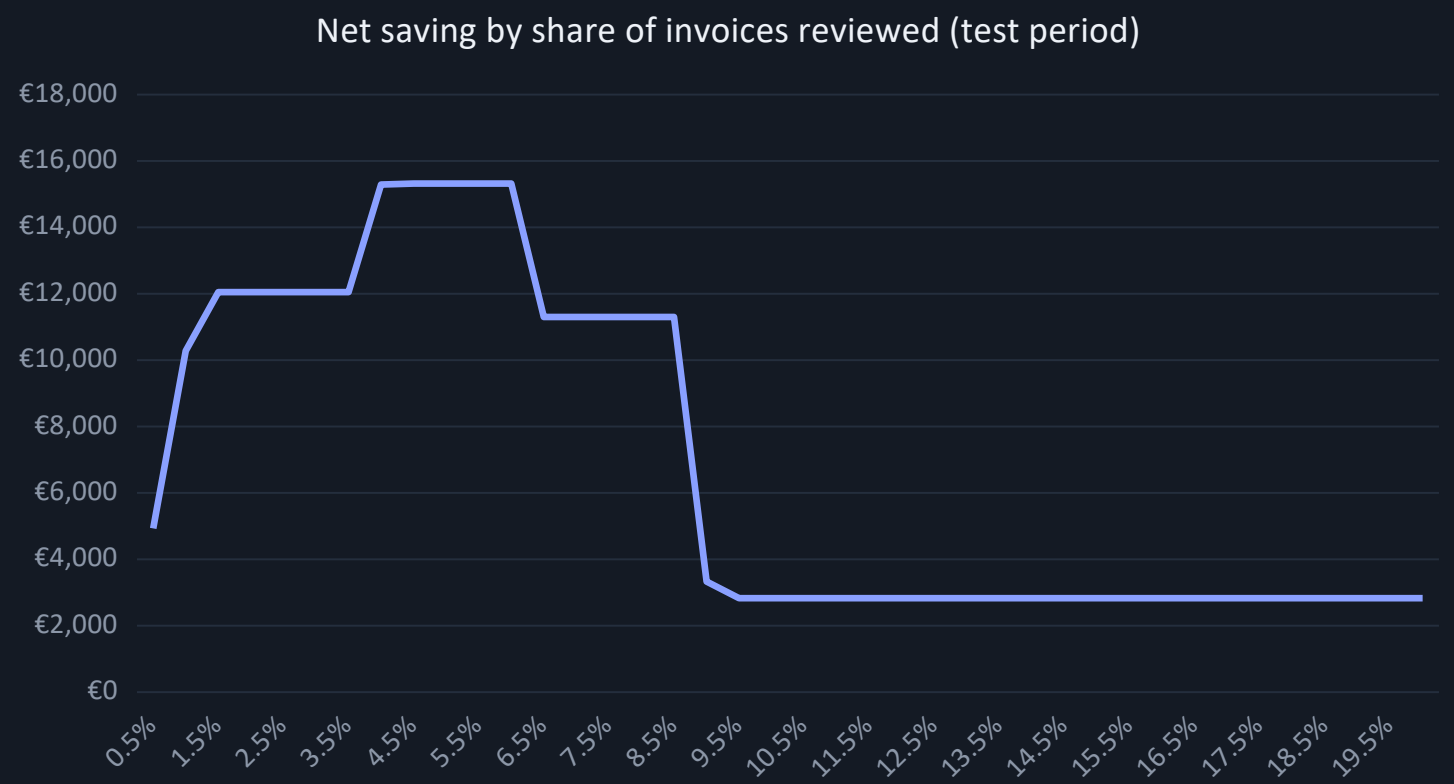
**79%**

next step correct, vs 76% for a Markov chain

*Tested on orders from a later period, with no future information in any prediction (unit-tested).*

FROM SCORE TO DECISION

# A good model isn't automatically a business case




**4.2%**

of invoices is the best share to review, not all of them

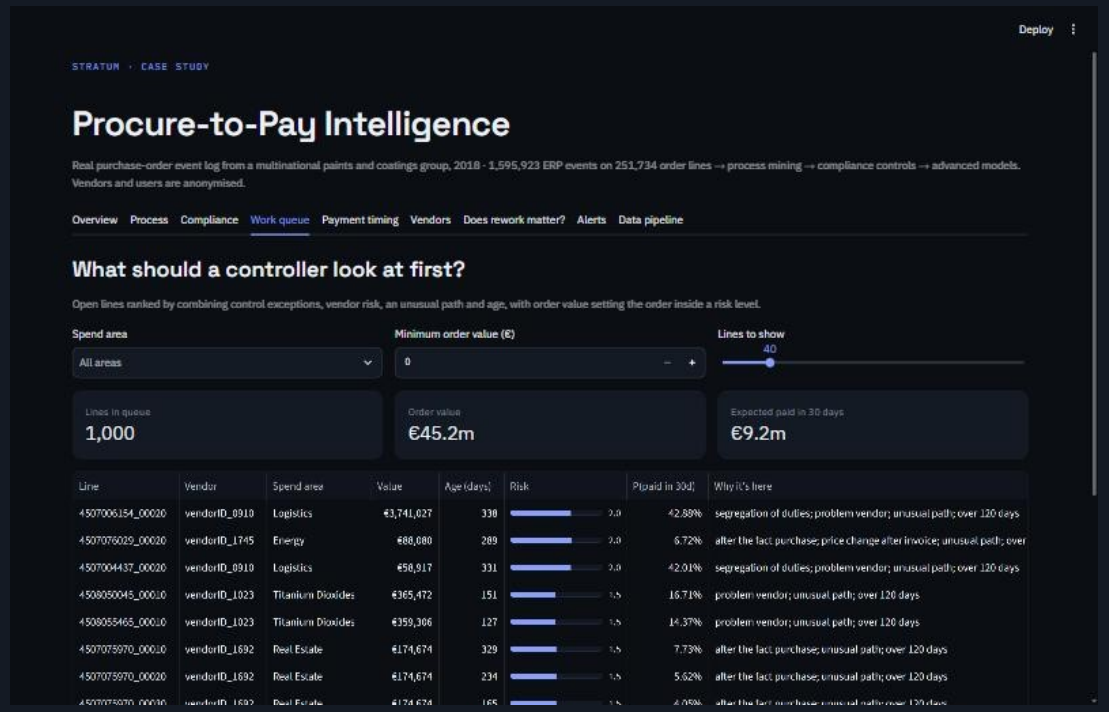
**34%**

of future corrections sit in that small slice

*ROC-AUC 0.75. Assumes €10 a review, €180 a correction and half prevented, which are inputs rather than findings. At €25 a review, no share pays.*

DELIVERY

# A work queue, not just a report




Work queue: open lines ranked, with the reason each one is there




Causal view: naive gap against the adjusted effect

Nine tabs: Overview · Process · Compliance · Work queue · Payment timing · Vendors · Rework · Alerts · Data pipeline

CONCLUSIONS

# What I'd tell the CFO

## Review the leaking controls

7,551 lines had one person on both sides of a control. Three other controls held fully: find out why, and copy it.

## Look at the wait after booking

That step holds the most waiting time. Check it against payment terms before changing anything.

## Don't chase order changes

Reducing rework is unlikely to speed payment. Spend that effort elsewhere, with Packaging the one area to look at.

## Run cash from the open book

A forecast built on line-level timing gives finance a 30-day view that's better than repeating last month.

HOW IT'S BUILT

# What makes this work trustworthy

### Tested forward in time

Every model is scored on later data, against something simple

### Reported what didn't work

Limits and weak spots are stated, not hidden

### Reproducible

One command from public download to dashboard, with identical results

### Tested

31 dbt tests plus 12 Python tests, including recovering known effects

TOOLKIT

Python · SQL · dbt · DuckDB · pm4py · lifelines · NumPyro · PyTorch · scikit-learn · SciPy · Plotly · Streamlit · pytest · GitHub Actions

STRATUM

# What's your ERP not telling you?

I turn the transaction history your systems already hold into clear answers on process, control and cash.

stratumdata.org

stratumfreelance@gmail.com